

Pooling Contracts between Venture Capital Institutions

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Abstract

Venture capital contracts are generally bargained deal by deal with entrepreneurs. However, contracts between accelerators and venture capitalists, and between venture capitalists and limited partners may impose blind commitment, include non-negotiable terms, shape as pooling contracts, etc. This article provides a theoretical framework for analyzing contracts between institutions in venture capital. If the principal reveals the expected outcome of the project after the investor's commitment, there will exist an optimal pooling mechanism, with a non-binding budget constraint for the principal in cash-rich economies. Otherwise, if the principal reveals the expected outcome of the project before the investor's commitment, the unique optimal mechanism will be pooling with a binding budget constraint for the principal in all economies.

JEL Codes: G24, G23, D82, D86

Keywords: Accelerators, Venture Capital, Limited Partners, Informed Principal, Mechanism Design, Blind-pool Contracts, Budget Constraints

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1 Introduction

Most of the contracts between venture capitalists (VCs) and their investors are blind pool contracts (Lerner et al., 2022) : they agree on the amount of committed capital, the investment split – typically 1-5% for VCs – and the return split – typically 20% for VCs. VCs’ investors, namely limited partners (LPs), have no say in the choice of each startup (Zeisberger et al., 2017). On the other hand, in general, accelerators – early-stage startup investors – propose a unique investment mechanism for startups and their terms are imposed to follow-on venture capitalists. Accelerators such as Y Combinator and Techstars run cohort-based programs of roughly three months that combine seed financing with a structured curriculum of training, mentorship, and investor networking, culminating in a “Demo Day” at which the cohort’s startups pitch to a large audience of follow-on investors. For example, Y Combinator invests \$500K per startup (\$125K post-money SAFE for 7% equity and \$375K uncapped MFN SAFE¹), non-negotiable. These contracts have in common that the terms are fixed ex-ante, but the second type of contracts goes further : a fixed amount of money will be invested on each startup by the first mover.

Despite a few spectacular failures (e.g., the New Chip Accelerator in Austin in 2023), there is evidence that accelerators may significantly increase venture fundraising and scale (Gonzalez-Uribe and Leatherbee, 2017). Conversely, VCs are more controversial since management fees – typically 2% of committed capital – may incentivize funds to grow at the expense of performance (Lerner and Nanda, 2020; Metrick and Yasuda, 2010). Thus, LPs’ blind commitment is suspected to be socially suboptimal (Lerner and Nanda, 2020). But, supposing limited liability, Axelson et al. (2009) argue that blind commitment minimizes the agency’s incentives to overstate projects’ values. Fang et al. (2015) confirms empirically that for the same institutions, co-investing in a deal by deal basis with funds yields worse results than blind pool deals. And while blind pool contracts are predominant, LPs’ demand for investing in VC is high and concentrated (Gourier et al., 2024; Lerner and Nanda, 2020). In fact, we now from Myerson (1983) that without loss of generality, a common mechanism can be proposed by an informed principal whatever the specific projects encountered. In this paper, we propose the first comprehensive analysis of the contracts between institutions in the venture capital market, using an informed principal framework with competitive agents (Maskin and Tirole, 1992; Myerson, 1983; Holmstrom and Myerson, 1983; Guesnerie and Laffont, 1984; Kamien and Schwartz, 1991).

Our theoretical model studies contracts in which the first mover (“the agency”) secures the followers’ (“the investors”) commitment to co-invest despite incentive frictions. When the principal reveals the expected outcome of the projects after the investors’ commitment (as in the relationship between VCs and LPs), investors may tolerate losses on individual projects as long as they break even on average; when

¹A Most-Favored Nation (MFN) SAFE is a convertible instrument that automatically adopts the most favorable terms obtained by any other investor between the date of issuance and the next priced equity round, ensuring the accelerator never invests on worse terms than subsequent investors.

the principal reveals the expected outcome of the projects before the investors' commitment (as in the relationship between accelerators and VCs), investors may refuse to lose money on any project. Our striking finding is that under either commitment regime, there are pooling contracts which are strategically optimal for the agency. In particular, under *ex-post commitment*—when investors require incentive-compatible deal-by-deal profitable contracts—the unique optimal contract is pooling: the agency ensures that the investors are confident in the fact that the agency is not incentivized to misvalue any project because it could yield a negative utility for the investors. Pooling contracts, combined with a binding budget constraint on the agency, ensure that every financed project yields positive utility to investors. A binding budget constraint for the agency ensures that the agency has skin in the game, so that committing its full budget always aligns its incentives with investing in the best projects. Under *ex-ante commitment*, the agency's budget constraint is not binding in general. Unsurprisingly, in this case, the agency extracts all the investors' surplus and, when possible, funds all the profitable projects. But under *ex-post commitment*, there are always some profitable projects which remain unfunded.

The rest of the paper proceeds as follows. Section 2 lays out the environment—agents, payoffs, and the informed-principal problem—and shows how pooling contracts arise as an immediate consequence of Myerson's inscrutability principle. As a point of reference, Section 3 solves the problem when investors share the agency's information. The heart of the analysis follows in Sections 5 and 4, which characterize the optimal contract under the two commitment regimes: under *ex-ante commitment*, the break-even condition holds only in expectation, whereas under *ex-post commitment*, investors walk away from any project that loses money on its own. Section 6 then asks how these conclusions depend on the aggregate supply of capital, contrasting cash-poor and cash-rich economies. Section 7 concludes, and all proofs are collected in the Appendix.

2 Model

2.1 Agents

Suppose there is no time preference and all the agents below are risk-neutral.

Let's introduce entrepreneurs without initial budget. Each entrepreneur carries a project which requires a sunk investment cost $I \in (0, 1)$ and yields a stochastic outcome in $\{0, 1\}$. 0 means failure and 1 means success. Without I , the project cannot succeed. An entrepreneur must hence collect at least I from outside sponsors, who can split the investment burden. A project is characterized by its probability of success $\pi \in [0, 1]$. Note that the probability of success is equal to the expected outcome. Each entrepreneur knows the expected outcome of his project. Let f be the probability density function of π , and let F denote its cumulative distribution

function.

Potential sponsors are investors with an aggregate budget $B_i \geq 0$. The investors know the distribution of π . In addition, they form a competitive market: they strategically underbid one another in order to secure investment opportunities. Throughout, we represent them by a price-taking representative investor, as is standard in the literature.

Financing can also be obtained from an agency with an initial budget $B_a \geq 0$. For each project, the agency knows the exact value of its expected outcome. An *investment allocation* σ specifies, for each project π , the agency's contribution to the investment cost I . The investor is expected to complete the remaining $\sigma_i(\pi)$. A *share allocation* ρ specifies, for each project π , the outcome $\rho(\pi)\pi$ obtained by the agency. The investor receives the remaining $\rho_i(\pi)\pi$. Formally:

$$\sigma_i(\pi) = \begin{cases} I - \sigma(\pi) & \text{if } \sigma(\pi) > 0 \\ 0 & \text{otherwise} \end{cases}$$

and:

$$\rho_i(\pi) = \begin{cases} 1 - \rho(\pi) & \text{if } \rho(\pi) > 0 \\ 0 & \text{otherwise} \end{cases}$$

The agency signals projects to investors by investing in. Thereby, we combine the certification and financing channels proposed by Howell (2017).

2.2 Definitions and assumptions

A project's *net present value* is defined as the difference of its expected outcome and its investment cost. For notational simplicity, the *market net present value*, which corresponds to the outside option's net present value, is taken to be 0. Then, a project is *worth financing* if its net present value is positive. A project is *financed* if it receives the required investment cost.

(A1): $\mathbb{E}(\pi) - I < 0 < 1 - I$ and f is log-concave

The first part of (A1) means that the average project is not worth financing but that the best project is worth financing. The second part—that $\log f$ is concave—is a standard regularity condition satisfied by the uniform, normal, logistic, and exponential distributions, among others. It implies that the hazard rate $f/(1 - F)$ is increasing.

(A2): $B_a < I(1 - F(I))$ and $B_a + B_i \geq I(1 - F(I))$

The first part of (A2) means that the agency cannot finance all worth financing projects alone: it needs outside investors. This is the economically relevant case—even the largest VCs and accelerators cannot finance all the projects alone. The second part means that together the agency and the investor can finance all projects.

(A3): The agency is profit-maximizing

The agency's payoff is:

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \quad \text{for each } \pi$$

The investors' payoff is:

$$U_i(\pi) = (1 - \rho(\pi))\pi - (I - \sigma(\pi)) \quad \text{for each } \pi$$

The investment strategies must be budget balanced both for the agency and the investor:

$$B_a \geq \int \sigma(\pi) f(\pi) d\pi \tag{BC_a}$$

$$B_i \geq \int \sigma_i(\pi) f(\pi) d\pi \tag{BC_i}$$

We have assumed here that the budgets B_a and B_i are common knowledge.

3 Benchmark

Before proceeding, it is useful to establish a benchmark result, namely that even without informational frictions, the first mover has a huge advantage. We will also introduce conceptual notions for the general resolution.

Game 1 (Benchmark).

$t = 0$ *Nature draws π*

$t = 1$ *The agency and the investors learn π*

$t = 2$ *The agency proposes $(\rho(\pi), \sigma(\pi))$*

$t = 3$ *Investors accept or refuse*

$t = 4$ *The agency executes $(\rho(\pi), \sigma(\pi))$*

$t = 5$ *Payoffs are realized*

Proposition 1. *Any budget balanced contract which finances all the worth financing projects by binding the investor's participation constraint on each project is optimal.*

Proof. See Appendix. □

Example 1. *Let $\pi \sim U[0, 1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$ (assumptions (A1)–(A3) hold). Take $\sigma^* = 0.1$ (agency's budget binding). Then $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ for $\pi \geq I$, increasing from $1/6$ at $\pi = I$ to $1/2$ at $\pi = 1$. The agency captures the full surplus $U_a(\pi) = \pi - I$ on every project; the investor earns zero rent.*

Indeed, since the agency is profit-maximizing, the investor's rationality constraint is binding for each project. Otherwise, since the investor market is competitive, the agency could contract with a cheaper investor, thereby making more profit. Thus, the agency's optimization problem directly reduces to a welfare maximization problem under budget constraints. This proposition means that when there is no asymmetric information, the agency extracts all the surplus from the investor project-wise, thanks to its first mover advantage.

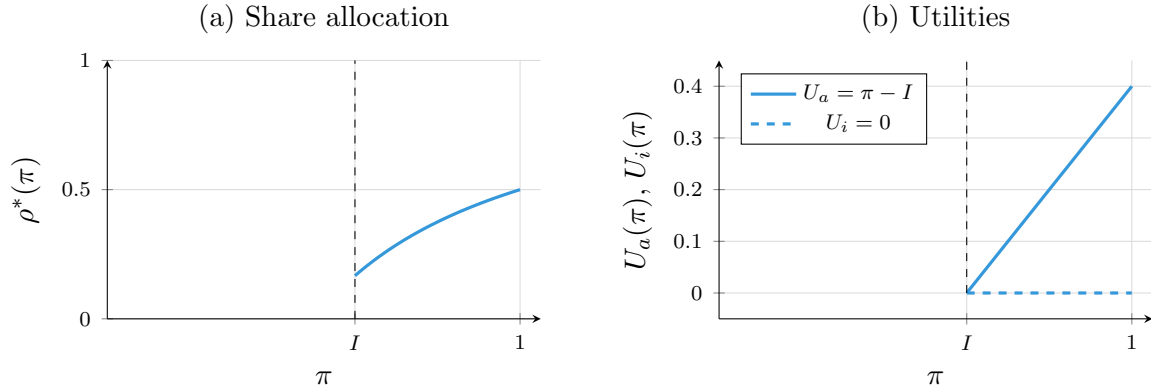


Figure 1: Informed investors under the benchmark. Parameters: $\pi \sim U[0, 1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in π . (b) The agency captures the full surplus $\pi - I$; the investor earns zero rent.

4 Ex-post commitment investors

Game 2 (Ex-post commitment).

- $t = 0$ Nature draws π
- $t = 1$ The agency learns π
- $t = 2$ The agency proposes (ρ, σ)
- $t = 3$ The agency reveals π
- $t = 4$ The investors accept or refuse
- $t = 5$ The agency executes $(\rho(\pi), \sigma(\pi))$
- $t = 6$ Payoffs are realized

This timing maps directly onto the relationship between an accelerator and the follow-on venture capitalists it solicits. At $t = 2$, the accelerator announces the cohort terms (uniform SAFE structure, fixed equity stake, fixed dollar investment per startup). At $t = 3$, the accelerator stages a Demo Day on which each startup is publicly presented to the follow-on investors: this is the moment at which π is revealed to the investors. At $t = 4$, the follow-on venture capitalists decide deal by deal whether to participate in the round, having observed the quality of each project. Investors retain the ability to walk away from any project whose value-per-dollar they judge insufficient, which is what the pointwise individual-rationality constraint formalizes below.

Under ex-post commitment investors, the individual-rationality constraint must hold pointwise, on every financed project:

$$\forall \pi, \quad (1 - \rho(\pi))\pi - (I - \sigma(\pi)) \geq 0 \quad (IR_{i,p})$$

Example 2 (Why terms that vary with the announcement unravel). *Four projects, $I = 1/2$ and $B_a = 0.4$: two of quality $\pi = 1/2$, one of $\pi = 3/4$, one of $\pi = 1$. Suppose the agency could write a separate contract for each and leave every investor exactly indifferent. Panel (a) of Table 1 does that: each investor holds $U_i = 0$, the agency collects 0.95, and only 0.25 of the budget is spent. Project by project the allocation satisfies $(IR_{i,p})$.*

It is not implementable, because the agency is the party that announces which project is which. Panel (b) has it report one of the $\pi = 1/2$ projects as $\pi = 1$ and take the contract written for $\pi = 1$: that pays $0.6 \times \frac{1}{2} - 0.1 = 0.20 > 0$, and the investor who accepted those terms holds $0.4 \times \frac{1}{2} - 0.4 = -0.20$. Its pointwise constraint fails. Nothing is special about this pair: every project below $\pi = 1$ prefers the terms written for $\pi = 1$, so no schedule that varies with the announcement survives the agency's own incentive constraint.

What survives is a single pair offered to every project. Panel (c) shows the best one, $(\rho, \sigma) = (0.6, 0.2)$: it finances the two projects with $\pi \geq 3/4$ and spends $2 \times 0.2 = B_a$. Both properties of Proposition 2 are visible here. The contract does not discriminate, and the budget is exhausted—raising σ further would finance no additional project, lowering it would leave the agency able to buy a larger share. The financing threshold sits at $3/4$, strictly above $I = 1/2$. The two projects at $\pi = 1/2$ are exactly marginal, so this example illustrates the threshold rather than the strict exclusion of Proposition 2(1); with a continuum of qualities, the projects in $[I, \pi_{\min}^*)$ are strictly positive-NPV and go unfinanced, which is what Example 3 computes.

Project	$\pi = 1/2$	$\pi = 1/2$	$\pi = 3/4$	$\pi = 1$
<i>(a) One contract per project: feasible pointwise</i>				
(ρ, σ)	(0.1, 0.05)	(0.1, 0.05)	(0.4, 0.05)	(0.6, 0.10)
Agency U_a	0	0	0.25	0.50
Investor U_i	0	0	0	0
<i>(b) The agency reports the second project as $\pi = 1$</i>				
(ρ, σ)	(0.1, 0.05)	(0.6, 0.10)	(0.4, 0.05)	(0.6, 0.10)
Agency U_a	0	0.20	0.25	0.50
Investor U_i	0	−0.20	0	0
<i>(c) One contract for all, and a saturated budget</i>				
(ρ, σ)	—	—	(0.6, 0.20)	(0.6, 0.20)
Agency U_a	—	—	0.25	0.40
Investor U_i	—	—	0	0.10

Table 1: Four projects, $I = 1/2$, $B_a = 0.4$. Panel (a) leaves every investor indifferent but cannot be implemented: in panel (b) the agency announces a $\pi = 1/2$ project as $\pi = 1$, gains 0.20, and pushes that investor to -0.20 . Panel (c) is the optimum of Proposition 2: one pair for everyone, the budget spent in full, and a rent left on the project above the margin.

The key result is that, despite the pointwise constraint, the optimal allocation remains pooling—and is fully characterized.

Proposition 2. *Under ex-post commitment investors, the optimal allocation is pooling. Any optimal allocation $(\rho^*, \sigma^*, \pi_{\min}^*)$ satisfies:*

1. *Some positive-NPV projects are excluded: $\pi_{\min}^* > I$.*
2. *The agency saturates its budget:*

$$\sigma^* = \frac{B_a}{\int_{\pi_{\min}^*}^1 f(\pi) d\pi}.$$

3. The share is pinned down by the threshold:

$$\rho^* = 1 - \frac{I - \sigma^*}{\pi_{\min}^*}.$$

4. The optimal threshold $\pi_{\min}^*$ satisfies the first-order condition:

$$(\pi_{\min}^* - I)f(\pi_{\min}^*) = \frac{\int_{\pi_{\min}^*}^1 (\pi - \pi_{\min}^*)f(\pi) d\pi}{\pi_{\min}^* \int_{\pi_{\min}^*}^1 f(\pi) d\pi} \cdot \sigma^* \cdot f(\pi_{\min}^*) + \frac{I - \sigma^*}{(\pi_{\min}^*)^2} \int_{\pi_{\min}^*}^1 \pi f(\pi) d\pi.$$

The right-hand side is strictly positive, forcing $\pi_{\min}^* > I$: the agency excludes some positive-NPV projects to limit the surplus transferred to investors.

Proof. See Appendix. □

Example 3. Same parameters as Example 1. For $f \equiv 1$ on $[0, 1]$, the first-order condition of Proposition 2 reduces to a cubic in $\pi_{\min}^*$ with a unique root $\pi_{\min}^* \approx 0.748 > I$ (uniqueness proved in Appendix). Projects with $\pi \in [I, \pi_{\min}^*)$ are positive-NPV but excluded. Then $\sigma^* \approx 0.397$ and $\rho^* \approx 0.729$. The investor earns a strictly positive rent $U_i(\pi) = 0.271\pi - 0.203 > 0$ on every financed project (zero only at the margin $\pi = \pi_{\min}^*$). Relative to the ex-ante commitment benchmark of Example 4, the agency finances fewer projects and leaves a rent on every deal.

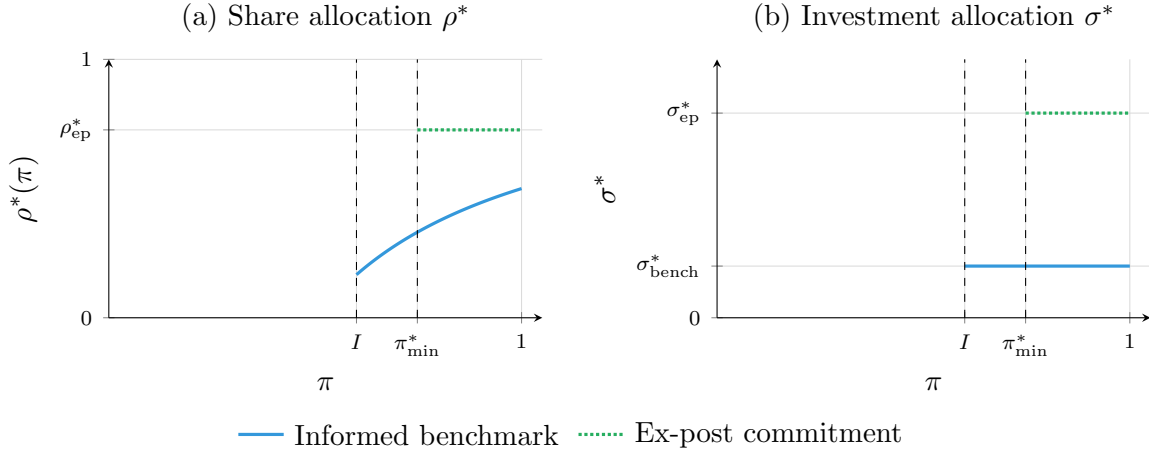


Figure 2: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0, 1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$. Under the informed benchmark, $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is tailored to each project. Under ex-post commitment, the optimal allocation is pooling and the threshold rises above I . The agency finances fewer projects but invests more per financed project.

The logic behind Proposition 2 mirrors that of Proposition 3, but with one crucial difference at each step. Because investors observe π before committing, $(IR_{i,p})$ must hold project by project: $(1 - \rho(\pi))\pi \geq I - \sigma(\pi)$ for all $\pi \geq \pi_{\min}^*$. Combined with

(IC_a), pointwise IR forces the optimal allocation to be pooling (Lemma 2): the agency cannot tailor ρ project by project without violating either IC or IR. The pooling share ρ^* is set so that the marginal project $\pi_{\min}^*$ just breaks even for the investor; every infra-marginal project $\pi > \pi_{\min}^*$ therefore yields a strictly positive rent $(1 - \rho^*)\pi - (I - \sigma^*) > 0$, which the agency cannot extract. To limit these rents, the agency voluntarily raises the threshold: rather than finance projects in $[I, \pi_{\min}^*)$ at the cost of transferring surplus on every deal, it excludes them. The optimal $\pi_{\min}^* > I$ balances the foregone NPV against the saved rent.

The binding budget constraint on the agency plays a complementary role: it gives the agency skin in the game, ensuring that committing its full budget aligns its private incentives with selecting the best projects.

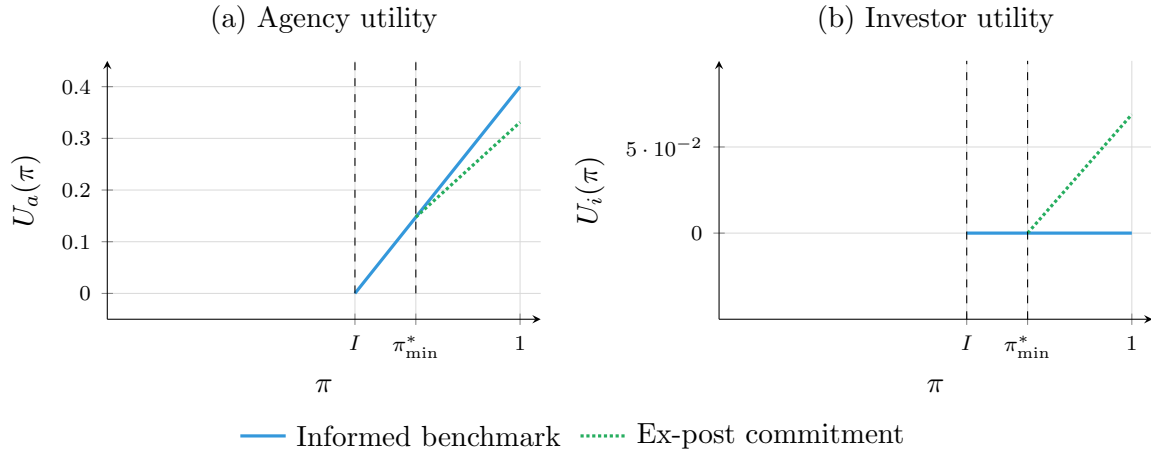


Figure 3: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters as Figure 2. (a) Agency utility: the benchmark captures $\pi - I$; ex-post commitment captures less on a narrower range. (b) Investor utility: zero under the benchmark, strictly positive for $\pi \geq \pi_{\min}^{\text{ep}}$ under ex-post commitment.

5 Ex-ante commitment investors

Game 3 (Ex-ante commitment).

$t = 0$ Nature draws π

$t = 1$ The agency learns π

$t = 2$ The agency proposes (ρ, σ)

$t = 3$ The investors accept or refuse

$t = 4$ The agency reveals π

$t = 5$ The agency executes $(\rho(\pi), \sigma(\pi))$

$t = 6$ Payoffs are realized

This timing maps onto the relationship between a venture capital fund and its limited partners. At $t = 2$, the fund proposes the terms of the partnership (committed capital, management fee, carried interest, investment policy). At $t = 3$, the LPs commit their capital before any specific project has been identified. The fund then sources and selects investments over the fund life; at $t = 4$, the agency reveals π by selecting a specific startup and issuing a capital call to its LPs to fund that investment. By that stage the LPs are already contractually committed and the participation constraint can only bind in expectation, as the formalization below makes explicit.

Under ex-ante commitment investors, the individual-rationality constraint binds only in expectation²:

$$\int [\rho_i(\pi)\pi - \sigma_i(\pi)] f(\pi) d\pi \geq 0 \quad (IR_i)$$

As in the benchmark, the agency extracts the full investor surplus. Restricting attention to pooling allocations yields a sharp characterization:

Proposition 3. *When investors are ex-ante commitment, a pooling allocation (ρ^*, σ^*) with financing threshold $\pi_{\min}$ is an optimal solution, characterized by:*

1. *The financing threshold is $\pi_{\min}^* = I$: all positive-NPV projects are financed.*
2. *The agency's share is:*

$$\rho^* = 1 - \frac{(1 - F(I))(I - \sigma^*)}{\int_I^1 \pi f(\pi) d\pi}$$

²Maskin and Tirole (1992)

3. The agency's investment share belongs to:

$$\sigma^* \in \left[I - \frac{B_i}{1 - F(I)}, \frac{B_a}{1 - F(I)} \right] \cap (0, I]$$

The contract form is not uniquely determined—any allocation satisfying IC and the binding (IR_i) is optimal—but the welfare level is. The pooling allocation with threshold I is one such solution. The cash-poor case $(B_a + B_i < I(1 - F(I)))$ is treated in the Appendix.

Proof. See Appendix. $\square$

Example 4. Same parameters as Example 1. The financing threshold is $\pi_{\min}^* = I = 0.6$: all positive-NPV projects are financed. Taking $\sigma^* = 0.1$ (agency's budget binding) yields $\rho^* = 0.375$. The agency's payoff per project is $U_a(\pi) = 0.375\pi - 0.1$; the investor's payoff is $U_i(\pi) = 0.625\pi - 0.5$, negative for $\pi < 0.8$ but zero in expectation over $[I, 1]$.

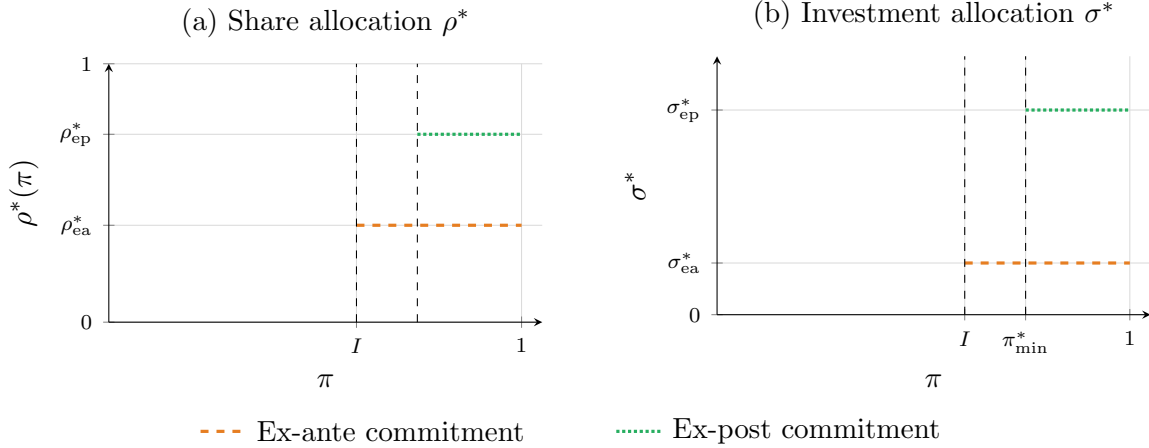


Figure 4: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as Figure 2. The ex-ante commitment regime finances all positive-NPV projects ($\pi_{\min}^* = I$); the ex-post commitment regime excludes the interval $[I, \pi_{\min}^{ep})$ and invests more per financed project.

The logic behind Proposition 3 runs as follows. Because the investor commits before π is revealed, the participation constraint (IR_i) is only required to hold in expectation. Under ex-ante commitment, (IR_i) binds in expectation: any slack would be competed away by the price-taking representative investor. Substituting the binding (IR_i) into the agency's objective collapses it to a pure welfare-maximization program, $\max \int_{\pi_{\min}}^1 (\pi - I) f(\pi) d\pi$. This objective coincides with the planner's: the agency finances the maximum number of positive-NPV projects within the available budget. The budget constraints then pin down the financing threshold and the investment split.

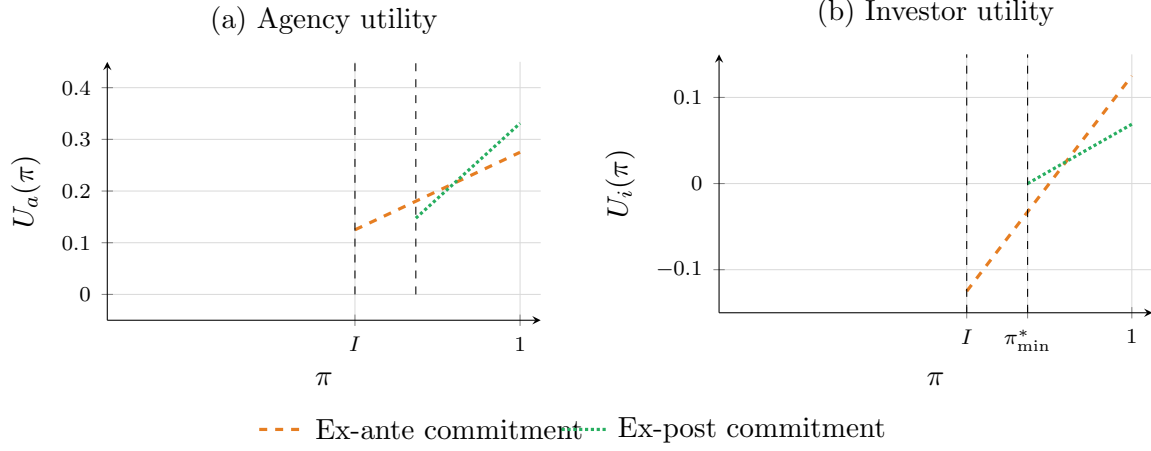


Figure 5: Utilities under the two commitment regimes. Same parameters. (a) Agency utility: the ex-ante commitment regime yields higher expected profit (wider financed range, starts at I). (b) Investor utility: zero *in expectation* under ex-ante commitment (negative for low- π projects, positive for high- π projects); strictly positive for every $\pi \geq \pi_{\min}^{\text{ep}}$ under ex-post commitment. The investor prefers ex-post commitment; the agency prefers ex-ante commitment.

6 Cash-poor vs. cash-rich economies

The analysis so far has assumed that aggregate budgets are sufficient to finance every positive-NPV project ((A2), second part). We now relax this assumption and characterize the full parameter space in the (B_a, B_i) plane. Let $\pi_{\min}^{\text{ep}}$ denote the threshold defined by Proposition 2 under assumption (A2); it depends on B_a but not on B_i .

Proposition 4. *In the (B_a, B_i) plane, three regions arise:*

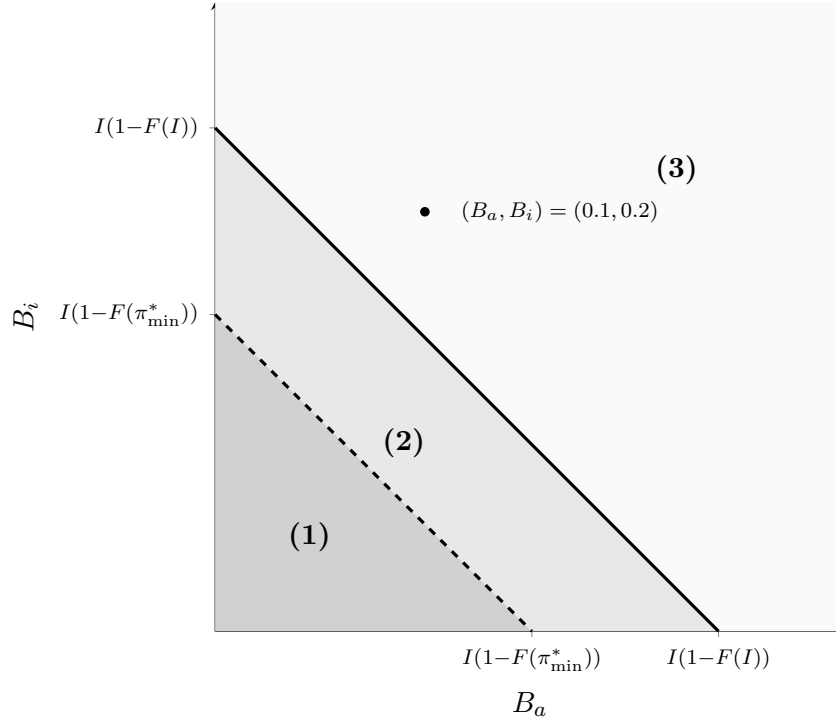
1. **Severely cash-poor** ($B_a + B_i < I(1 - F(\pi_{\min}^{\text{ep}}))$): both regimes coincide. Both budget constraints bind, and the threshold is determined by the aggregate budget alone:

$$\pi_{\min}^* = (1 - F)^{-1}\left(\frac{B_a + B_i}{I}\right).$$

The budget constraint, not the informational friction, pins down the allocation.

2. **Moderately cash-poor** ($I(1 - F(\pi_{\min}^{\text{ep}})) \leq B_a + B_i < I(1 - F(I))$): the ex-ante commitment regime remains constrained-efficient (it finances every project above the budget-constrained threshold), while the ex-post commitment regime introduces an additional inefficiency: the informational friction raises the threshold further above the budget-constrained level.
3. **Cash-rich** ($B_a + B_i \geq I(1 - F(I))$): this is assumption (A3). The ex-ante commitment regime finances all positive-NPV projects ($\pi_{\min}^* = I$), while the ex-post commitment regime excludes the interval $[I, \pi_{\min}^{\text{ep}})$.

Figure 6 maps these three regions. The key takeaway is that the informational friction is *slack* in the severely cash-poor region (1), where both regimes deliver the same constrained-efficient allocation, and *binds* in regions (2) and (3), where the ex-post commitment regime sacrifices positive-NPV projects that the ex-ante commitment regime would finance.



- (1) Severely cash-poor: both regimes coincide; only the budget binds
- (2) Moderately cash-poor: ex-ante commitment efficient, ex-post commitment inefficient
- (3) Cash-rich: ex-ante commitment efficient, ex-post commitment inefficient

Figure 6: Budget regions in the (B_a, B_i) plane (f uniform, $I = 0.6$, $\pi_{\min}^{\text{ep}} \approx 0.748$). Solid line: $B_a + B_i = I(1 - F(I))$. Dashed line: $B_a + B_i = I(1 - F(\pi_{\min}^{\text{ep}}))$. In region (1) the budget constraint dominates and both commitment regimes yield the same allocation. In regions (2) and (3) the ex-post commitment regime introduces an inefficiency that the ex-ante commitment regime avoids.

7 Conclusion

7.1 Pointwise participation as a covenant

The two games differ in when investors commit, and the participation constraint differs with them—pointwise under ex-post commitment, in expectation under ex-ante commitment. Read as a clause rather than as a consequence of timing, the pointwise constraint is available to an investor that commits early.

Fix a pooling allocation (ρ, σ) with threshold $\pi_{\min}$. The investor's payoff $U_i(\pi) = (1 - \rho)\pi - (I - \sigma)$ is affine and increasing in π , so

$$U_i(\pi) \geq 0 \text{ on every financed project} \iff \rho \leq 1 - \frac{I - \sigma}{\pi_{\min}},$$

the condition Proposition 2 holds with equality. The whole family of pointwise constraints collapses into one inequality on the announced terms, and an investor can check it at $t = 2$, before any project exists. What the constraint bears on is the schedule $(\rho, \sigma, \pi_{\min})$, and π enters it only through the threshold the agency announces.

Enforcement comes from the budget. The agency saturates B_a at a per-project contribution $\sigma = B_a/(1 - F(\pi_{\min}))$, so the measure of projects it can carry on the announced terms is exactly $1 - F(\pi_{\min})$: the budget fixes how many deals there are. Which deals they are is settled by the agency's own payoff $U_a(\pi) = \rho\pi - \sigma$, increasing in π , so an agency holding a fixed number of slots fills them from the top. The threshold is self-enforcing, and an investor that observes neither π nor the selection obtains the ex-post allocation out of a blind pool.

Two things follow. Demo Day and the covenant implement the same constraint by different means—one buys it with information, the other with contract—and Proposition 2 characterizes both. And the covenant is privately valuable and socially costly in one movement: an investor bound in expectation alone earns zero, while under the covenant it earns a strictly positive rent on every financed project (Example 3), and the projects in $[I, \pi_{\min}^*)$ are what pays for that rent.

What separates the two regimes is then bargaining power rather than timing. The investors of Section 2 underbid one another, so an investor demanding the covenant is replaced by one accepting the constraint in expectation, and the covenant is not an equilibrium of the ex-ante game as written. Sustaining it asks that investors be scarce enough relative to agencies for a refusal to be credible, and the market structure that delivers that scarcity—a monopolist limited partner, an oligopoly with capacity limits, a repeated relationship—is left open here. Gompers and Lerner (1996) document the counterpart in partnership agreements: limited partners obtain covenants restricting a fund's investment behaviour, and covenant intensity moves with fundraising conditions, which is the observable form of the scarcity the argument requires.

The covenant also need not be bargained for. Investors that manage other people's money operate under mandates written on positions rather than on portfolios: the prudent person principle of Solvency II (Directive 2009/138/EC, Article 132) requires an insurer to identify, measure and monitor the risk of each asset it holds, a pension fund's investment policy statement fixes admissibility criteria an individual holding must meet, and fiduciary standards are drafted over investments rather than over their average. Such an investor does not decline the expectation-form contract, it is unable to sign it—and inability is what a competitor cannot undercut. The veto the covenant requires is then supplied from outside the financing relationship, which is the one source of it that leaves the right to propose with the agency.

Read this way, prudential regulation of the investor becomes a determinant of which projects are financed. Where a regime assesses holdings one at a time, the investors of Section 5 behave as covenant holders, the threshold is $\pi_{\min}^*$ and the projects in $[I, \pi_{\min}^*)$ go unfunded; where a regime lets an investor be judged on its portfolio, the constraint binds in expectation, the threshold is I , and every positive-NPV project is financed. The direction is worth stating plainly: a rule that protects the investor position by position transfers a rent to it and withdraws capital from projects that are worth financing.

The comparison is not hypothetical: the venture capital industry ran it. Through 1978, ERISA's prudent man rule was read as assessing a pension fiduciary's holdings one at a time, and pension funds stayed out of venture capital because a single investment in a start-up could be judged imprudent on its own. In early 1979 the Department of Labor ruled that portfolio diversification enters the prudence of an individual investment (Gompers and Lerner, 1998). That is the move from $(IR_{i,p})$ to (IR_i) , carried out by a regulator rather than negotiated in a contract. Commitments to venture capital rose, with the effect concentrated on pension funds—the investors the rule bound—and statistically distinguishable from the effect on taxable investors.

What was financed moved with how much was financed, and in the direction Proposition 2 implies. Gompers and Lerner (1998) find that venture organizations became more likely to raise funds with an *early-stage* focus after the clarification, and read that shift as following from investments being judged by their contribution to portfolio risk rather than by their own. In the model this is the threshold falling from $\pi_{\min}^*$ to I , and the projects financing reaches once the constraint binds in expectation are exactly those of $[I, \pi_{\min}^*)$ —the ones a pointwise constraint leaves out. The reading they advance for the composition of funds is what the pointwise constraint formalizes, and the model delivers the volume and the composition from the same threshold.

The 1979 clarification relaxed the constraint; a contract can instead satisfy it. A position-level standard leaves an insurer or a pension fund free to hold venture capital provided every position carries a non-negative expected payoff, and Proposition 2 describes a contract that delivers exactly that: a uniform amount per project, a share capped at $1 - (I - \sigma)/\pi_{\min}$, and an agency budget exhausted at the announced number of deals. Those three terms are ones a limited partnership agreement can carry—a disclosed capital envelope, a co-investment fixed per deal rather than as a percentage of the fund, and uniform terms across the portfolio—and together they bound the investor's payoff below by zero on every financed project while leaving π unobserved.

Two routes therefore serve one objective, and the model prices them against each other. Relaxing the standard moves the investor to (IR_i) : the threshold falls to I , every positive-NPV project is financed, and the investor earns zero. Meeting the standard by covenant holds it at $(IR_{i,p})$: it earns a rent on every position, and the projects of $[I, \pi_{\min}^*)$ go unfunded. A regulator seeking to move insurance capital towards young firms chooses between more firms financed by capital that had to be freed and fewer firms financed by capital that did not.

The argument has two boundaries. The covenant bears on the qualitative standard—the prudent person principle—rather than on the capital requirement, which the standard formula assigns by asset class and not by the terms of a contract. And what it bounds is the expected payoff conditional on π : the outcome of a single project remains in $\{0, 1\}$, so the guarantee is one of valuation at the point of investment rather than of realized return.

The two readings are live at the same time. Where a mandate carries explicit per-asset requirements, as the prudent person principle of Solvency II does in Europe, the contract of Proposition 2 is what lets an investor hold venture capital at all; where prudence has been assessed on the portfolio since 1979, as it has in the United States under ERISA, the ex-ante game remains the description. The same institutional pair therefore sits in one game or the other according to the jurisdiction binding its limited partners, which is a second reason to read the two commitment regimes as contractual rather than as fixed attributes of a relationship.

7.2 Limitations

Our results rest on several modelling choices worth acknowledging. First, the informational monopoly assumption—that the agency knows π exactly and is the unique informed party—is stronger than the empirical record. Yet the VC industry is highly concentrated: Lerner and Nanda (2020) document that a small number of top-tier firms capture most of the sector’s returns. An oligopoly of informed intermediaries is the relevant benchmark, and extending the analysis to that case would not alter the main insights. Second, we assume no externalities across projects; in practice, financing one startup may crowd out capital from another or generate network spillovers. Third, the investment cost I is constant across projects; letting $I(\pi)$ vary with quality could shift the optimal threshold. Fourth, the framework abstracts from the bargaining dynamics between entrepreneurs and investors—a dimension illustrated by the evolution of accelerator contracts away from uniform cohort terms (e.g., Y Combinator). Fifth, we do not derive the mapping between commitment regimes and institutional pairs from primitives. The paper takes as given that the limited-partner-to-general-partner relationship operates under ex-ante commitment and that the accelerator-to-venture-capitalist relationship operates under ex-post commitment, and motivates each mapping through the natural correspondence between Demo Day and the revelation of π in the ex-post timing, and between the capital call and the revelation of π in the ex-ante timing. A model of why these particular commitment regimes co-evolved with these particular institutional pairs—e.g., through fund-life horizons, regulatory constraints on LPs, or repeated-game considerations between accelerators and follow-on VCs—is left for future work. These extensions we leave for future work.

7.3 Implications for practice and policy

The central result of the paper is the joint characterization of the unique optimal contract under ex-post commitment: it is pooling *and* it saturates the agency’s budget. Relaxing either property violates incentive compatibility on the agency side or pointwise individual rationality on the investor side. The New Chip Accelerator in Austin, which collapsed in 2023, illustrates the failure mode the model predicts when the disciplining device is absent. New Chip ran cohorts of dozens of startups simultaneously without committing meaningful capital alongside investors, and without a binding budget constraint that would have forced selection.

The paper also identifies an efficiency cost attached to a transition the industry has already made. Ewens et al. (2018) document that the fall in the cost of experimentation moved early-stage investors toward smaller initial stakes in more startups, and with it toward a sequence in which an accelerator or seed intermediary stands between the entrepreneur and the venture capitalist. That sequence is the ex-post commitment timing of Section 4: quality is revealed on Demo Day, before the follow-on investor commits. Proposition 2 says what such a timing costs. Because the investor may walk away project by project, the agency’s individual-rationality constraint binds pointwise; the optimal contract pools, saturates the budget, and sets a financing threshold strictly above I . Projects with $\pi \in [I, \pi_{\min}^*)$ are positive-NPV and go unfinanced. The same projects are financed under the ex-ante commitment timing of Section 5, where the investor commits before quality is revealed. The move to an accelerator-intermediated world is therefore not neutral: it carries a selection stage whose incentives exclude projects that the earlier arrangement funded.

This gives the state a lever, and Section 6 says where it bites. In a cash-rich economy the binding constraint is not the supply of capital but the pointwise constraint of the party that decides last, so public money spent on one more accelerator adds capacity to the stage that generates the exclusion. Public money committed *before* quality is revealed does not: it enters the relationship whose incentives are already aligned, where the budget constraint is slack and every positive-NPV project is financed. The institutional form that corresponds to it is the limited partner. A state acting as an LP in venture funds—the European Investment Fund and Bpifrance both operate this way—supplies capital under ex-ante commitment, which is the timing Proposition 3 shows to be free of the exclusion. Acting as a venture capitalist would place the state on the informed side of the same relationship and change nothing about the ordering that creates the cost.

The advantage disappears in one region only, and for a reason internal to the model rather than a caveat added to it. In a severely cash-poor economy the constraint that binds is the shortage of capital itself: the threshold is set by the aggregate budget, the informational friction is slack, and the two timings deliver the same allocation (Proposition 4, region 1). Public money then finances the same projects whether it enters through an accelerator or as a limited partner, and what matters is how much

of it arrives. In a moderately cash-poor economy the friction binds again, and ex-ante commitment keeps its advantage.

Either role presupposes that the state can execute it, which is a separate question from the one this model answers, and one on which the record is mixed (Lerner, 2022).

A further takeaway follows from reading the pointwise constraint as a covenant. The agency's selectivity rests on two quantities that require no knowledge of any project: its budget, which is disclosed, and its contribution per project, which is a contractual parameter. Fixing σ at $B_a/(1 - F(\pi_{\min}))$ caps at $1 - F(\pi_{\min})$ the measure of deals the agency can carry, and its payoff $\rho\pi - \sigma$ being increasing in π settles the rest: an agency holding a fixed number of slots fills them from the top. A co-investment requirement is therefore an instrument of selection, operable by an investor or a regulator that observes neither project quality nor the agency's choices.

Two features separate it from the retention rules written for securitization, where an originator keeps a slice of an issue to make its effort or its private signal credible (Chemla and Hennessy, 2014). Here no party exerts effort and no signal is transmitted: the retained slice exhausts a budget, and selection follows from a capacity limit rather than from incentives. And the instrument carries a price. The rent it secures for the financier is paid by the projects in $[I, \pi_{\min}^*)$, so a regulator reaching for it buys protection for the investor with financing for the firm.

8 Appendix

8.1 Proof of Proposition 1 (informed investors).

Lemma 1. *No project with $\pi < I$ is financed in any optimal mechanism.*

Proof. If $\pi < I$, then $\pi - I < 0$: the project has negative expected net value. The investor's payoff from financing it is $(1 - \rho(\pi))\pi - (I - \sigma(\pi)) \leq \pi - I < 0$ whenever $\sigma(\pi) \leq \rho(\pi)\pi$ (i.e. the agency does not subsidize the investor beyond the project's return). Thus $(IR_{i,p})$ cannot be satisfied unless the agency transfers more than the project generates, which would also make the agency's payoff negative. Since both parties lose, no rational mechanism finances projects below I . $\square$

Step 1: The investor's rationality constraint binds

Since the agency is profit-maximizing, the investor's rationality constraint is binding for each project. Otherwise, since the investor market is ex-ante commitment, the agency could contract with a cheaper investor, thereby making more profit:

$$1 - \rho^*(\pi) > \frac{I - \sigma^*(\pi)}{\pi} \Rightarrow \exists \rho^\#(\pi) \in \left(\rho^*(\pi), 1 - \frac{I - \sigma^*(\pi)}{\pi} \right),$$

$$\int_{\pi_{\min}^*}^1 (\rho^*(\pi)\pi - \sigma^*(\pi))f(\pi) d\pi < \int_{\pi_{\min}^*}^1 (\rho^\#(\pi)\pi - \sigma^*(\pi))f(\pi) d\pi$$

and $(\sigma^*, \rho^\#)$ satisfies (BC_a) , (BC_i) , and (IR_i) , a contradiction. Thus:

$$\rho^*(\pi) = 1 - \frac{I - \sigma^*(\pi)}{\pi}$$

Step 2: Reducing the agency's optimization problem

The agency's optimization problem reduces to:

$$\max_{(\pi_{\min}, \sigma, \rho)} \int_{\pi_{\min}}^1 \left(\left(1 - \frac{I - \sigma(\pi)}{\pi} \right) \pi - \sigma(\pi) \right) f(\pi) d\pi$$

$$= \max_{(\pi_{\min}, \sigma, \rho)} \int_{\pi_{\min}}^1 (\pi - I) f(\pi) d\pi$$

and:

$$\rho(\pi) = 1 - \frac{I - \sigma(\pi)}{\pi}$$

8.2 Proof of Proposition 3 (ex-ante commitment investors).

Step 1: The investor's rationality constraint binds in expectation.

Since the agency is profit-maximizing, (IR_i) binds in expectation. Otherwise, the agency could reduce the investor's share and strictly increase its profit: if (IR_i) were slack, there would exist $\hat{\rho}$ such that $(\sigma^*, \hat{\rho})$ remains feasible and yields a higher agency payoff, a contradiction. Hence:

$$(1 - \rho^*) \frac{\int_{\pi_{\min}^*}^1 \pi f(\pi) d\pi}{\int_{\pi_{\min}^*}^1 f(\pi) d\pi} = I - \sigma^*.$$

Step 2: Reducing the optimization problem.

With the binding (IR_i) , the agency's objective becomes

$$\max_{\sigma, \pi_{\min}} \int_{\pi_{\min}}^1 \left(\left(1 - \frac{(I - \sigma)(1 - F(\pi_{\min}))}{\int_{\pi_{\min}}^1 \pi f(\pi) d\pi} \right) \pi - \sigma \right) f(\pi) d\pi$$

under (BC_a) and (BC_i) . Any allocation satisfying IC and the binding (IR_i) delivers the same welfare, so the contract form is not uniquely determined.

Step 3: First-order conditions.

Expanding the derivative with respect to $\pi_{\min}^*$ by the Leibniz rule and the quotient rule yields

$$\frac{\partial \mathbb{E}[\pi \mid \pi \geq \pi_{\min}]}{\partial \pi_{\min}} = \frac{f(\pi_{\min}) \left(\int_{\pi_{\min}}^1 \pi f(\pi) d\pi - \pi_{\min} (1 - F(\pi_{\min})) \right)}{(1 - F(\pi_{\min}))^2}.$$

After substitution and simplification, the first-order condition on $\pi_{\min}^*$ reduces to $(-\pi_{\min}^* + I)f(\pi_{\min}^*) = 0$. Since $f(\pi_{\min}^*) > 0$, we obtain $\pi_{\min}^* = I$: all positive-NPV projects are financed.

The first-order condition with respect to σ^* reduces to the identity $\mathbb{E}[\pi \mid \pi \geq \pi_{\min}^*] = \int_{\pi_{\min}^*}^1 \pi f(\pi) d\pi / (1 - F(\pi_{\min}^*))$, which is true by definition. Hence σ^* is not pinned down by the unconstrained first-order conditions; it is determined by the budget constraints.

Step 4: Budget case analysis under (A3).

The Lagrangian

$$L = - \int_{\pi_{\min}}^1 \left(\left(1 - \frac{(I - \sigma)(1 - F(\pi_{\min}))}{\int_{\pi_{\min}}^1 \pi f(\pi) d\pi} \right) \pi - \sigma \right) f d\pi + \lambda_1 \left(\sigma \int_{\pi_{\min}}^1 f - B_a \right) + \lambda_2 \left((I - \sigma) \int_{\pi_{\min}}^1 f - B_i \right)$$

gives the stationarity conditions

$$\frac{\partial L}{\partial \sigma} = (-\lambda_1 + \lambda_2)(1 - F(\pi_{\min})) = 0, \quad \frac{\partial L}{\partial \pi_{\min}} = ((\pi_{\min} - I) - \lambda_1 \sigma - \lambda_2(I - \sigma))f(\pi_{\min}) = 0.$$

Hence $\lambda_1^* = \lambda_2^*$ and $\pi_{\min}^* = I + \lambda_1^* I$. Under (A3), $B_a + B_i \geq I(1 - F(I))$, so both multipliers vanish, $\pi_{\min}^* = I$, and the investment share satisfies

$$\sigma^* \in \left] \frac{B_a}{1 - F(I)}, I - \frac{B_i}{1 - F(I)} \right[\cap (0, I].$$

Any allocation in this interval satisfying IC and the binding (IR_i) is optimal. $\square$

8.3 Proof of Proposition 2 (ex-post commitment investors).

Part I: pooling is optimal.

We first restate the incentive-compatibility constraint as a pair of differential conditions that will be used as the state equations of the optimal control problem.

Lemma 2. *The agency's incentive-compatibility constraint is equivalent to its first- and second-order conditions:*

$$(IC_a) \Leftrightarrow \begin{cases} \frac{d\sigma(\pi)}{d\pi} = \pi \frac{d\rho(\pi)}{d\pi} & (IC_a^{(1)}) \\ \frac{d\rho(\pi)}{d\pi} \geq 0 & (IC_a^{(2)}) \end{cases}$$

Proof. ($\Rightarrow$) The first-order condition is

$$\frac{\partial V}{\partial \hat{\pi}}(\pi, \pi) = 0 \Leftrightarrow \frac{d\sigma(\pi)}{d\pi} = \pi \frac{d\rho(\pi)}{d\pi}.$$

Differentiating,

$$\frac{d^2\sigma(\pi)}{d\pi^2} = \frac{d\rho(\pi)}{d\pi} + \pi \frac{d^2\rho(\pi)}{d\pi^2},$$

and the second-order condition $\frac{d^2\sigma(\pi)}{d\pi^2} \geq \pi \frac{d^2\rho(\pi)}{d\pi^2}$ yields $\frac{d\rho(\pi)}{d\pi} \geq 0$ by substitution.

($\Leftarrow$) Consider

$$\frac{\partial V}{\partial \hat{\pi}}(\pi, \hat{\pi}) = \frac{\partial(\pi\rho(\hat{\pi}))}{\partial \rho} \frac{d\rho(\hat{\pi})}{d\pi} - \frac{d\sigma(\hat{\pi})}{d\pi}.$$

Writing the first-order condition at $\hat{\pi}$ gives $\frac{\partial(\hat{\pi}\rho(\hat{\pi}))}{\partial \rho} \frac{d\rho(\hat{\pi})}{d\pi} = \frac{d\sigma(\hat{\pi})}{d\pi}$, hence

$$\frac{\partial V(\pi, \hat{\pi})}{\partial \hat{\pi}} = \left(\frac{\partial(\pi\rho(\hat{\pi}))}{\partial \rho} - \frac{\partial(\hat{\pi}\rho(\hat{\pi}))}{\partial \rho} \right) \frac{d\rho(\hat{\pi})}{d\pi}.$$

By the Mean Value Theorem, the sign of the right-hand side is that of $\frac{\partial^2(\pi^*\rho(\hat{\pi}))}{\partial \rho \partial \pi}(\pi - \hat{\pi}) \frac{d\rho(\hat{\pi})}{d\pi}$ for some π^* between π and $\hat{\pi}$. Since $\frac{\partial^2(\pi^*\rho(\hat{\pi}))}{\partial \rho \partial \pi} = 1$, the second-order condition implies that $\hat{\pi} \mapsto V(\pi, \hat{\pi})$ is increasing until $\hat{\pi} = \pi$ and decreasing afterwards, which concludes the proof. $\square$

Using Lemma 2, the agency's program can be rewritten as the following optimal control problem:

$$\max_{(\pi_{\min}, \sigma, \rho)} \int_{\pi_{\min}}^1 (\rho(\pi)\pi - \sigma(\pi))f(\pi) d\pi$$

under:

$$\begin{aligned} B_a - \int_{\pi_{\min}}^1 \sigma(\pi)f(\pi) d\pi &\geq 0 & (BC_a) \\ B_i - \int_{\pi_{\min}}^1 (I - \sigma(\pi))f(\pi) d\pi &\geq 0 & (BC_i) \\ (1 - \rho(\pi))\pi - (I - \sigma(\pi)) &\geq 0 & (IR_{i,p}) \\ \rho'(\pi) = u(\pi), \quad \sigma'(\pi) = \pi u(\pi) & & (IC_a^{(1)}) \\ u(\pi) &\geq 0 & (IC_a^{(2)}) \\ \sigma(\pi_{\min}), \rho(\pi_{\min}), \sigma(1), \rho(1) &\text{ free} & (LC) \end{aligned}$$

Step 1: Applying Pontryagin's Maximum Principle

The Hamiltonian is:

$$\begin{aligned} H(\pi) = & (\rho(\pi)\pi - (1 + (\lambda_a - \lambda_i))\sigma(\pi))f(\pi) \\ & + \lambda_1(\pi)u(\pi) + \lambda_2(\pi)\pi u(\pi) \\ & + \eta_1(\pi)u(\pi) + \eta_2(\pi)(\pi - I - \rho(\pi)\pi + \sigma(\pi)) \end{aligned}$$

- Optimality condition

$$\frac{\partial H}{\partial u} = 0 \Leftrightarrow \lambda_1(\pi) + \lambda_2(\pi)\pi + \eta_1(\pi) = 0$$

- Multiplier conditions

$$\begin{aligned} \lambda_1'(\pi) &= -\frac{\partial H}{\partial \rho} = \eta_2(\pi)\pi - \pi f(\pi) \\ \lambda_2'(\pi) &= -\frac{\partial H}{\partial \sigma} = (1 + (\lambda_a - \lambda_i))f(\pi) - \eta_2(\pi) \end{aligned}$$

- Complementarity slackness conditions

$$\begin{aligned} \lambda_a &\geq 0, \lambda_i \geq 0, \eta_1(\pi) \geq 0, \eta_2(\pi) \geq 0 \\ \lambda_a \left(B_a - \int_{\pi_{\min}}^1 \sigma(\pi)f(\pi) d\pi \right) &= 0 \\ \lambda_i \left(B_i - \int_{\pi_{\min}}^1 (I - \sigma(\pi))f(\pi) d\pi \right) &= 0 \\ \eta_1(\pi)u(\pi) &= 0 \\ \eta_2(\pi)(\pi - I - \rho(\pi)\pi + \sigma(\pi)) &= 0 \end{aligned}$$

- Transversality conditions

$$\lambda_1(\pi_{\min}) = 0, \lambda_2(\pi_{\min}) = 0, \lambda_1(1) = 0, \lambda_2(1) = 0$$

Step 2: Rearranging the optimality condition and establishing $\lambda_a > \lambda_i$

From the optimality condition on σ in the Hamiltonian, maximizing H with respect to the allocation between σ and $I - \sigma$ yields the first-order condition $\lambda_a - \lambda_i = \frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi) d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi) d\pi} > 0$. The inequality holds because $\pi > \pi_{\min}$ on the integration domain and under (A2) the agency cannot self-finance. Hence $\lambda_a > \lambda_i$.

Now, from the optimality and transversality conditions:

$$\begin{aligned} -\eta_1(\pi) &= \lambda_1(\pi) + \lambda_2(\pi)\pi \\ &= \int_{\pi_{\min}}^{\pi} \lambda'_1(v) dv + \left(\int_{\pi_{\min}}^{\pi} \lambda'_2(v) dv \right) \pi \\ &= \int_{\pi_{\min}}^{\pi} (\eta_2(v)v - vf(v) + \pi((1 + (\lambda_a - \lambda_i))f(v) - \eta_2(v))) dv \\ &= \int_{\pi_{\min}}^{\pi} (\eta_2(v)v - \pi\eta_2(v)) dv - \int_{\pi_{\min}}^{\pi} vf(v) dv + (1 + (\lambda_a - \lambda_i))\pi(F(\pi) - F(\pi_{\min})) \end{aligned}$$

By the *Fundamental Theorem of Calculus*:

$$\begin{aligned} \pi(F(\pi) - F(\pi_{\min})) &= \int_{\pi_{\min}}^{\pi} \frac{d(v(F(v) - F(\pi_{\min})))}{dv} dv \\ &= \int_{\pi_{\min}}^{\pi} (vf(v) + (F(v) - F(\pi_{\min}))) dv \end{aligned}$$

Thus:

$$\begin{aligned} -\eta_1(\pi) &= \int_{\pi_{\min}}^{\pi} \eta_2(v)(v - \pi) dv - \int_{\pi_{\min}}^{\pi} vf(v) dv + (1 + (\lambda_a - \lambda_i)) \int_{\pi_{\min}}^{\pi} (vf(v) + (F(v) - F(\pi_{\min}))) dv \\ &= \int_{\pi_{\min}}^{\pi} \eta_2(v)(v - \pi) dv + (1 + (\lambda_a - \lambda_i)) \int_{\pi_{\min}}^{\pi} (F(v) - F(\pi_{\min})) dv + (\lambda_a - \lambda_i) \int_{\pi_{\min}}^{\pi} vf(v) dv \end{aligned}$$

Step 3: Showing $u = 0$

Suppose there exists an interval $\mathcal{I}$ on which $u(\pi) > 0$. By complementarity, $\eta_1(\pi) = 0$ on $\mathcal{I}$, so the expression derived in Step 2 gives:

$$\int_{\pi_{\min}}^{\pi} \eta_2(v)(v - \pi) dv + (1 + (\lambda_a - \lambda_i)) \int_{\pi_{\min}}^{\pi} (F(v) - F(\pi_{\min})) dv + (\lambda_a - \lambda_i) \int_{\pi_{\min}}^{\pi} vf(v) dv = 0.$$

Differentiating twice with respect to π yields $\eta_2(\pi) = (1 + 2(\lambda_a - \lambda_i))f(\pi) + (\lambda_a - \lambda_i)\pi f'(\pi)$.

We now use the *transversality conditions* at $\pi = 1$: $\lambda_1(1) = \lambda_2(1) = 0$. Integrating the multiplier equations *from* π *to* 1:

$$\lambda_1(\pi) = \int_{\pi}^1 (vf(v) - \eta_2(v)v) dv, \quad \lambda_2(\pi) = \int_{\pi}^1 (\eta_2(v) - (1 + (\lambda_a - \lambda_i))f(v)) dv.$$

Hence $\eta_1(\pi) = -\lambda_1(\pi) - \lambda_2(\pi)\pi = \int_{\pi}^1 \eta_2(v)(v - \pi) dv + (1 + (\lambda_a - \lambda_i))\pi \int_{\pi}^1 f(v) dv - \int_{\pi}^1 vf(v) dv \geq 0$ for $\pi < 1$, consistently with the complementarity condition.

Case A: $\eta_2(\pi) = 0$ **on a sub-interval** $[\pi_0, \pi_1] \subset \mathcal{I}$. Then the expression $\eta_2(\pi) = (1 + 2(\lambda_a - \lambda_i))f(\pi) + (\lambda_a - \lambda_i)\pi f'(\pi) = 0$ is an ODE whose solution is $f(\pi) = C\pi^{-(2+1/(\lambda_a - \lambda_i))}$ on $[\pi_0, \pi_1]$. Since $\lambda_a > \lambda_i$ (established in Step 2 above), the exponent satisfies $-(2 + 1/(\lambda_a - \lambda_i)) < -3$. A log-concave density f satisfies $(\log f)'' \leq 0$, i.e. $f''f - (f')^2 \leq 0$. For $f(\pi) = C\pi^{\gamma}$ with $\gamma < -3$, we have $(\log f)'' = -\gamma/\pi^2 > 0$, which violates log-concavity. This yields a contradiction whenever f is log-concave (a condition that holds for the uniform, normal, exponential, and most standard distributions).

Case B: $\eta_2(\pi) > 0$ **on** $\mathcal{I}$. By complementarity, $(1 - \rho(\pi))\pi - (I - \sigma(\pi)) = 0$ on $\mathcal{I}$. Differentiating with respect to π : $-\rho'(\pi)\pi + (1 - \rho(\pi)) + \sigma'(\pi) = 0$. Substituting $\sigma'(\pi) = \pi u(\pi) = \pi \rho'(\pi)$ from $(IC_a^{(1)})$ gives $1 - \rho(\pi) = 0$, so $\rho(\pi) = 1$ and $\sigma(\pi) = I$, implying $u(\pi) = \rho'(\pi) = 0$ —a contradiction with $u > 0$.

In both cases we reach a contradiction. We conclude that $u(\pi) = 0$ for all $\pi \geq \pi_{\min}$, which establishes that $\rho' = \sigma' = 0$, i.e. the optimal allocation is pooling.

Part II: threshold, investment, and share.

Step 1: The investor's rationality constraint binds pointwise.

Since the agency is profit-maximizing, $(IR_{i,p})$ binds at the margin. Otherwise, there would exist $\rho^{\#}$ with a strictly higher agency share for which the contract remains feasible:

$$1 - \rho^* > \frac{I - \sigma^*}{\pi_{\mu}^*} \Rightarrow \exists \rho^{\#} \in \left(\rho^*, 1 - \frac{I - \sigma^*}{\pi_{\mu}^*} \right),$$

$$\int_{\pi_{\min}^*}^1 (\rho^* \pi - \sigma^*) f(\pi) d\pi < \int_{\pi_{\min}^*}^1 (\rho^{\#} \pi - \sigma^*) f(\pi) d\pi$$

and $(\sigma^*, \rho^{\#})$ satisfies (BC_a) , (BC_i) , and $(IR_{i,p})$. Contradiction. Thus:

$$\rho^* = 1 - \frac{I - \sigma^*}{\pi_{\mu}^*}$$

Step 2: Reducing the agency's optimization problem

The agency solves:

$$\begin{aligned} & \max_{\pi_{\min}, \sigma} \int_{\pi_{\min}}^1 \left(\left(1 - \frac{I - \sigma}{\pi_{\min}} \right) \pi - \sigma \right) f(\pi) d\pi \\ &= \max_{\pi_{\min}, \sigma} \int_{\pi_{\min}}^1 \left(\pi - \frac{I\pi}{\pi_{\min}} + \frac{\sigma\pi}{\pi_{\min}} - \sigma \right) f(\pi) d\pi \end{aligned}$$

under (BC_a) and (BC_i) .

The Lagrangian is:

$$L = \int_{\pi_{\min}}^1 \left(\pi - \frac{I\pi}{\pi_{\min}} + \frac{\sigma\pi}{\pi_{\min}} - \sigma \right) f(\pi) d\pi + \lambda_1 \left(B_a - \sigma \int_{\pi_{\min}}^1 f(\pi) d\pi \right) + \lambda_2 \left(B_i - (I - \sigma) \int_{\pi_{\min}}^1 f(\pi) d\pi \right)$$

$$\frac{\partial L}{\partial \sigma} = 0 \Leftrightarrow \int_{\pi_{\min}}^1 \left(\frac{\pi}{\pi_{\min}} - 1 \right) f(\pi) d\pi - \lambda_1 \int_{\pi_{\min}}^1 f(\pi) d\pi + \lambda_2 \int_{\pi_{\min}}^1 f(\pi) d\pi = 0$$

$$\frac{\partial L}{\partial \sigma} = 0 \Leftrightarrow \lambda_1 - \lambda_2 = \frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi) d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi) d\pi} > 0$$

Thus $\lambda_1 > 0$ and $B_a = \sigma^* \int_{\pi_{\min}^*}^1 f(\pi) d\pi$.

$$\begin{aligned} \frac{\partial L}{\partial \pi_{\min}} = 0 & \Leftrightarrow -(\pi_{\min} - I)f(\pi_{\min}) + \int_{\pi_{\min}}^1 \pi(\sigma - I) \frac{\partial}{\partial \pi_{\min}} \left(\frac{1}{\pi_{\min}} \right) f(\pi) d\pi + \lambda_1 \sigma f(\pi_{\min}) + \lambda_2 (I - \sigma) f(\pi_{\min}) \\ & \Leftrightarrow -(\pi_{\min} - I)f(\pi_{\min}) + (\lambda_1 - \lambda_2) \sigma f(\pi_{\min}) + \lambda_2 I f(\pi_{\min}) + \frac{I - \sigma}{\pi_{\min}^2} \int_{\pi_{\min}}^1 \pi f(\pi) d\pi = 0 \\ & \Leftrightarrow (\pi_{\min} - I)f(\pi_{\min}) = \\ & \left(\frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi) d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi) d\pi} \right) \left(\frac{B_a}{\int_{\pi_{\min}}^1 f(\pi) d\pi} \right) f(\pi_{\min}) + \lambda_2 I f(\pi_{\min}) + \frac{I - \frac{B_a}{\int_{\pi_{\min}}^1 f(\pi) d\pi}}{\pi_{\min}^2} \int_{\pi_{\min}}^1 \pi f(\pi) d\pi \end{aligned}$$

Thus $\pi_{\mu}^* > I$.

Case 1 (cash-rich, under (A3)): $B_i > (I - \sigma^*) \int_{\pi_{\min}^*}^1 f(\pi) d\pi$, i.e. $\pi_{\min}^* > (1 - F)^{-1} \left(\frac{B_a + B_i}{I} \right)$.

Under assumption (A3), the total budget is sufficient to finance all positive-NPV projects, so the investor's budget constraint (BC_i) is slack: $\lambda_2 = 0$. This is the relevant case.

The first-order condition on $\pi_{\min}$ becomes:

$$(\pi_{\min} - I)f(\pi_{\min}) - \left(\frac{\int_{\pi_{\min}}^1 (\pi - \pi_{\min}) f(\pi) d\pi}{\pi_{\min} \int_{\pi_{\min}}^1 f(\pi) d\pi} \right) \left(\frac{B_a}{\int_{\pi_{\min}}^1 f(\pi) d\pi} \right) f(\pi_{\min}) - \frac{I - \frac{B_a}{\int_{\pi_{\min}}^1 f(\pi) d\pi}}{\pi_{\min}^2} \int_{\pi_{\min}}^1 \pi f(\pi) d\pi = 0$$

Since the agency's reduced objective $V(\pi_{\min})$ is continuous on the compact set $[I, \bar{p}]$ (where $\bar{p}$ is the largest feasible threshold satisfying $\sigma^* = B_a / \bar{F}(\pi_{\min}) \leq I$), it attains a maximum. At any interior maximum $\pi_{\min}^*$, the first-order condition holds. The right-hand side is strictly positive, so $\pi_{\min}^* > I$. The optimal allocation is then $\sigma^* = B_a / \bar{F}(\pi_{\min}^*)$ and $\rho^* = 1 - (I - \sigma^*) / \pi_{\min}^*$.

Case 2 (cash-poor boundary): $\lambda_2 = 0$ and $B_i = (I - \sigma^*) \int_{\pi_{\min}^*}^1 f(\pi) d\pi$.

This is the boundary of the cash-poor regime. Here the unconstrained optimum coincides with the budget-constrained threshold: $\pi_{\mu}^* = (1 - F)^{-1}(\frac{B_a + B_i}{I})$.

Case 3 (cash-poor): $\lambda_2 > 0$ and $B_i = (I - \sigma^*) \int_{\pi_{\min}^*}^1 f(\pi) d\pi$.

Both budget constraints bind. Thus:

$$\pi_{\mu}^* = (1 - F)^{-1}\left(\frac{B_a + B_i}{I}\right), \quad \sigma^* = \frac{B_a}{B_a + B_i} I$$

The condition $\lambda_2 > 0$ imposes:

$$\begin{aligned} & (\pi_{\min}^* - I) f(\pi_{\min}^*) \\ & - \left(\frac{\int_{\pi_{\min}^*}^1 (\pi - \pi_{\min}^*) f(\pi) d\pi}{\pi_{\min}^* \int_{\pi_{\min}^*}^1 f(\pi) d\pi} \right) \left(\frac{B_a}{\int_{\pi_{\min}^*}^1 f(\pi) d\pi} \right) f(\pi_{\min}^*) \\ & - \left(\frac{I - \frac{B_a}{\int_{\pi_{\min}^*}^1 f(\pi) d\pi}}{(\pi_{\min}^*)^2} \right) \int_{\pi_{\min}^*}^1 \pi f(\pi) d\pi \\ & = \lambda_2 I f(\pi_{\min}^*) > 0. \end{aligned}$$

The left-hand side is strictly positive, meaning the budget-constrained threshold $(1 - F)^{-1}(\frac{B_a + B_i}{I})$ is *above* the unconstrained optimum from Case 1. The binding budget forces the agency to finance even fewer projects than the informational friction alone would dictate.

8.4 Proof of Proposition 4 (cash-poor vs. cash-rich).

The Lagrangian of the ex-ante commitment program (see proof of Proposition 3) has stationarity conditions $\lambda_1^* = \lambda_2^*$ and $\pi_{\min}^* = I + \lambda_1^* I$. Three cases exhaust the parameter space.

Region (1): severely cash-poor. When $B_a + B_i < I(1 - F(\pi_{\min}^{\text{ep}}))$, both budget constraints bind in both regimes. Hence

$$\pi_{\min}^* = (1 - F)^{-1}\left(\frac{B_a + B_i}{I}\right), \quad \sigma^* = \frac{B_a}{B_a + B_i} I,$$

and $\rho^* = 1 - (I - \sigma^*)/\pi_{\min}^*$ in both regimes. Since the ex-post commitment first-order condition of Proposition 2 would require $\pi_{\min}^{\text{ep}}$ strictly above this budget-constrained threshold, the budget dominates and both regimes coincide.

Region (2): moderately cash-poor. When $I(1 - F(\pi_{\min}^{\text{ep}})) \leq B_a + B_i < I(1 - F(I))$, the budget-constrained threshold $\pi_{\min}^{\text{bdg}} := (1 - F)^{-1}((B_a + B_i)/I)$ lies in $[\pi_{\min}^{\text{ep}}, I)$. Under ex-ante commitment investors, the constrained-efficient allocation finances every project above $\pi_{\min}^{\text{bdg}}$. Under ex-post commitment investors, the informational friction raises the threshold to $\pi_{\min}^{\text{ep}} > \pi_{\min}^{\text{bdg}}$, excluding the interval $[\pi_{\min}^{\text{bdg}}, \pi_{\min}^{\text{ep}})$.

Region (3): cash-rich. When $B_a + B_i \geq I(1 - F(I))$ —that is, assumption (A3)—the ex-ante commitment threshold drops to $\pi_{\min}^* = I$ (Proposition 3), while the ex-post commitment threshold remains $\pi_{\min}^{\text{ep}} > I$ (Proposition 2). The two regimes diverge on the interval $[I, \pi_{\min}^{\text{ep}})$. $\square$

8.5 Uniqueness of the ex-post commitment threshold under f uniform.

For $f \equiv 1$ on $[0, 1]$: $1 - F(p) = 1 - p$ and $\int_p^1 \pi d\pi = (1 - p^2)/2$. Substituting $\sigma^* = B_a/(1 - p)$ and $\rho^* = 1 - (I - \sigma^*)/p$ into the agency's payoff $\int_p^1 (\rho^* \pi - \sigma^*) f(\pi) d\pi$ gives

$$V(p) = \frac{1 - p}{2p} [p^2 + p(1 - I) - I + B_a].$$

Differentiating,

$$V'(p) = \frac{-2p^3 + Ip^2 + I - B_a}{2p^2}, \quad V''(p) = -\frac{p^3 + I - B_a}{p^3}.$$

Since $p > I > 0$ and $B_a < I$ from (A2), $p^3 + I - B_a > 0$, so $V''(p) < 0$ for all $p > I$. The objective is strictly concave. Setting $V'(p) = 0$ gives the unique optimal threshold as the solution of $2p^3 - Ip^2 - (I - B_a) = 0$ in $(I, 1)$. $\square$

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